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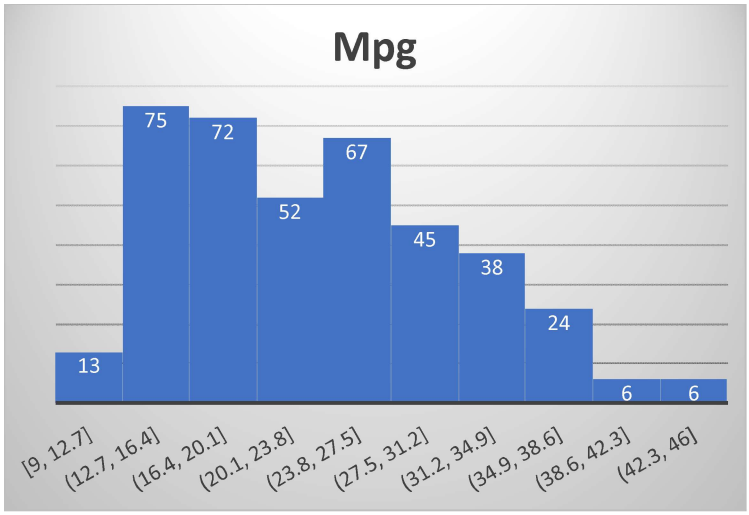
# Model for MPG dataset

**Since target variable is of continues and numeric also since there is more than one IV variable we need to use multiple linear regression.**

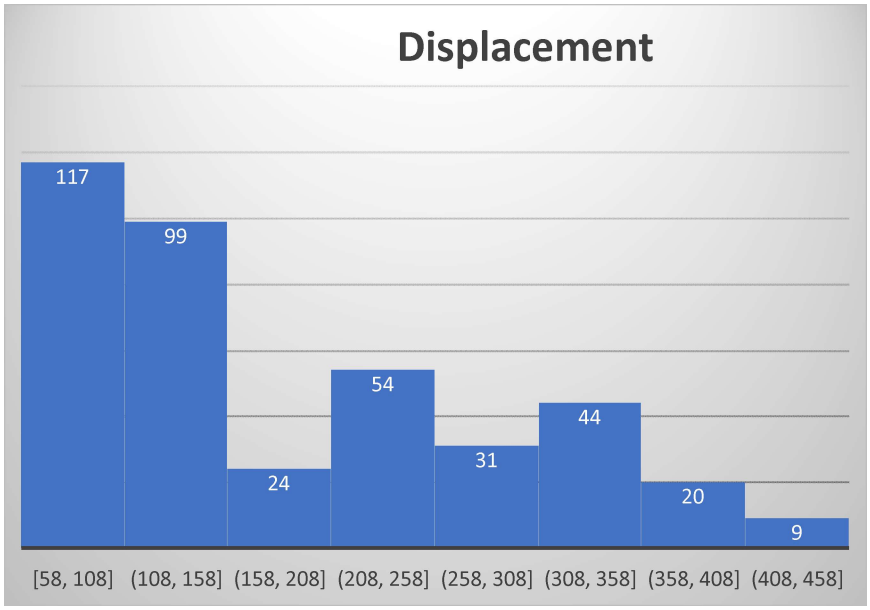
## Data Distribution of attributes

Mpg



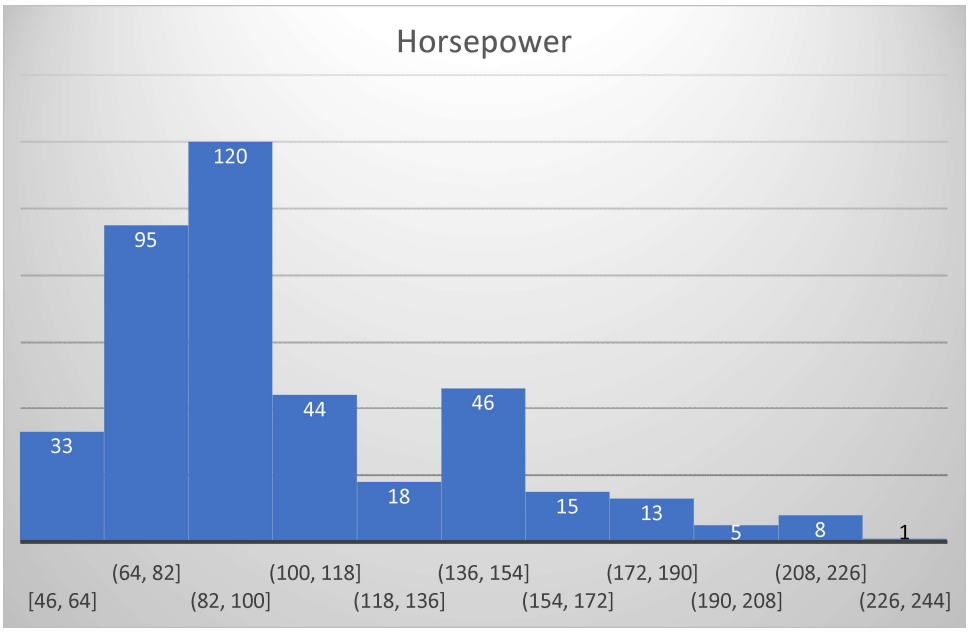

| *mpg* | |
|---|---|
| Mean | 23.51457 |
| Standard Error | 0.39178 |
| Median | 23 |
| Mode | 13 |
| Standard Deviation | 7.815984 |
| Sample Variance | 61.08961 |
| Kurtosis | -0.51078 |
| Skewness | 0.457066 |
| Range | 37.6 |
| Minimum | 9 |
| Maximum | 46.6 |
| Sum | 9358.8 |
| Count | 398 |

Displacement



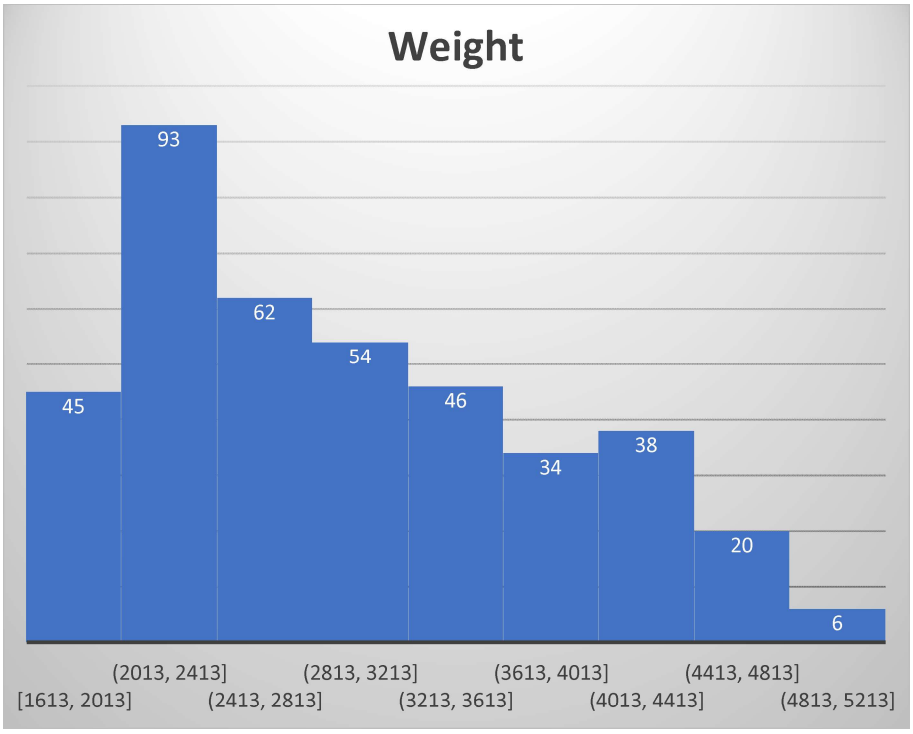

| *displacement* | |
|---|---|
| Mean | 193.4259 |
| Standard Error | 5.226575 |
| Median | 148.5 |
| Mode | 97 |
| Standard Deviation | 104.2698 |
| Sample Variance | 10872.2 |
| Kurtosis | -0.7466 |
| Skewness | 0.719645 |
| Range | 387 |
| Minimum | 68 |
| Maximum | 455 |
| Sum | 76983.5 |
| Count | 398 |

Horsepower



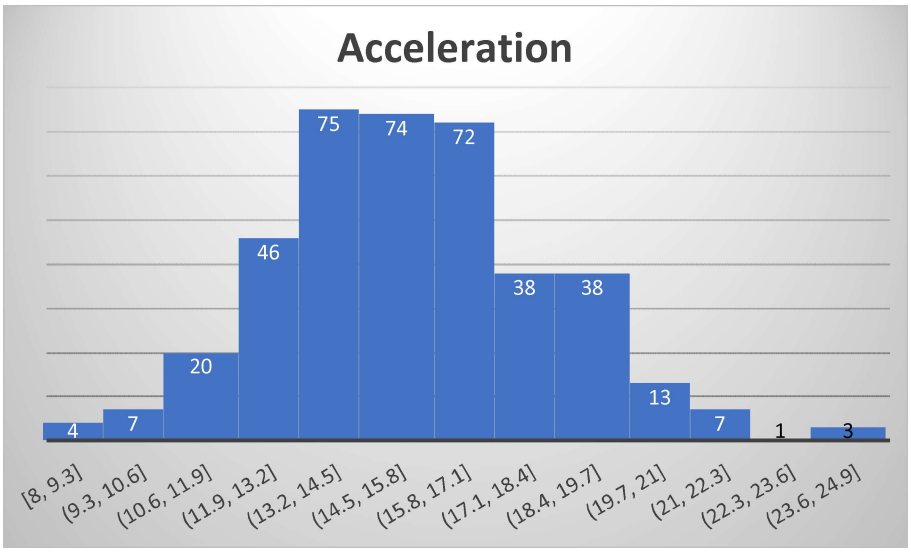

| *horsepower* | |
|---|---|
| Mean | 103.696 |
| Standard Error | 1.940386 |
| Median | 92 |
| Mode | 150 |
| Standard Deviation | 38.71058 |
| Sample Variance | 1498.509 |
| Kurtosis | 0.690832 |
| Skewness | 1.077186 |
| Range | 184 |
| Minimum | 46 |
| Maximum | 230 |
| Sum | 41271 |
| Count | 398 |

Weight




| *weight* | |
|---|---|
| Mean | 2970.425 |
| Standard Error | 42.44834 |
| Median | 2803.5 |
| Mode | 2130 |
| Standard Deviation | 846.8418 |
| Sample Variance | 717141 |
| Kurtosis | -0.78553 |
| Skewness | 0.531063 |
| Range | 3527 |
| Minimum | 1613 |
| Maximum | 5140 |
| Sum | 1182229 |
| Count | 398 |

Acceleration

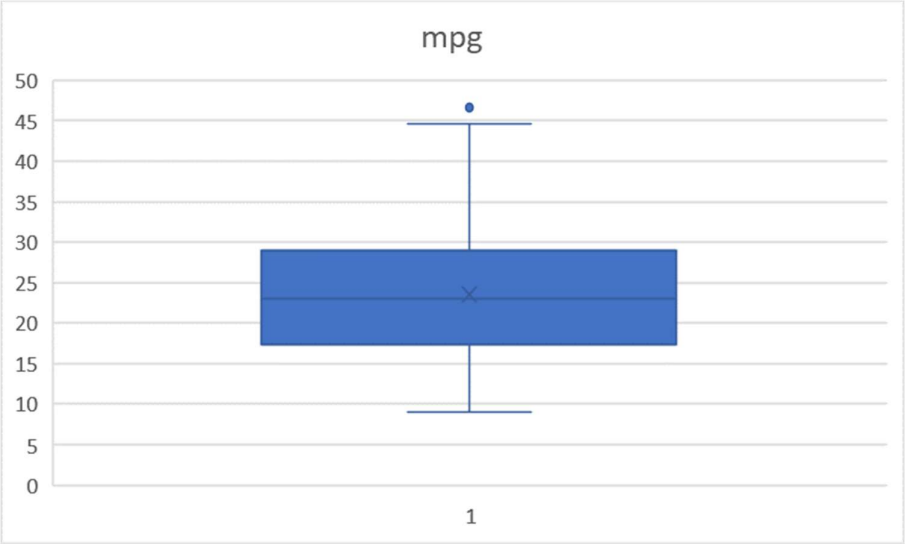



| *acceleration* | |
|---|---|
| Mean | 15.56809 |
| Standard Error | 0.13823 |
| Median | 15.5 |
| Mode | 14.5 |
| Standard Deviation | 2.757689 |
| Sample Variance | 7.604848 |
| Kurtosis | 0.419497 |
| Skewness | 0.278777 |
| Range | 16.8 |
| Minimum | 8 |
| Maximum | 24.8 |
| Sum | 6196.1 |
| Count | 398 |

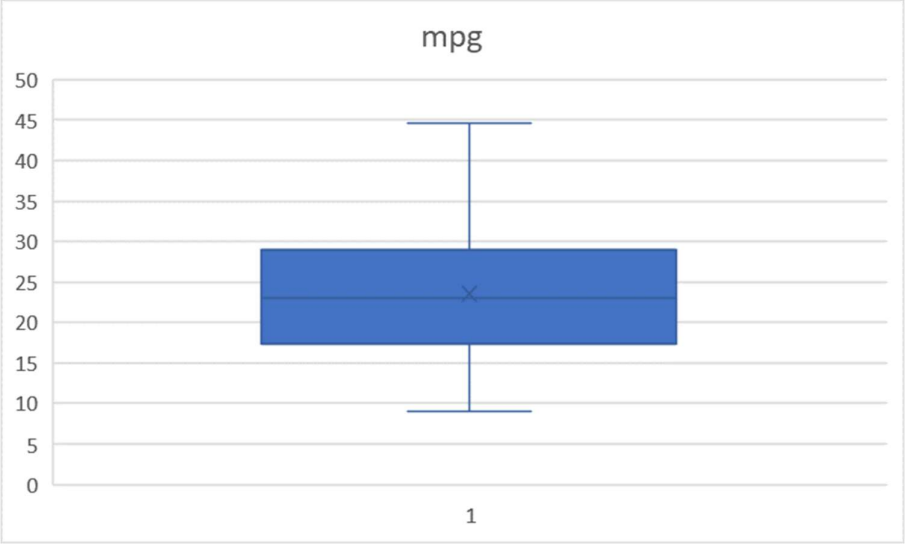
**Outlier handling for attributes**

**For below numeric variables**

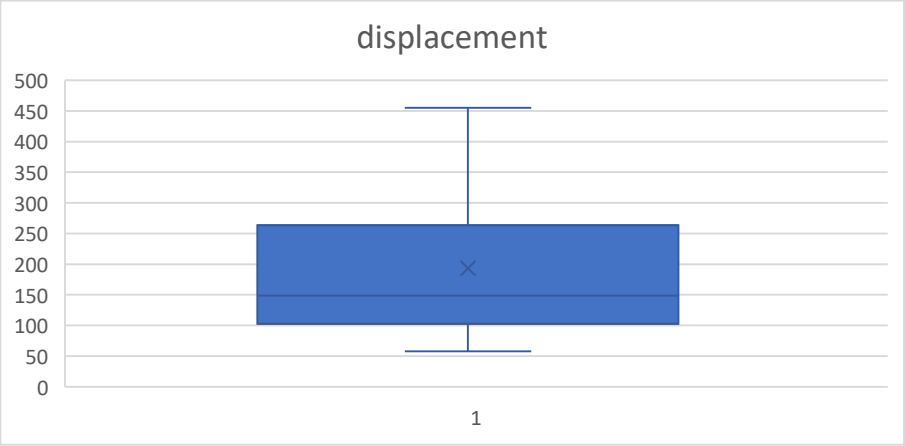
Mpg



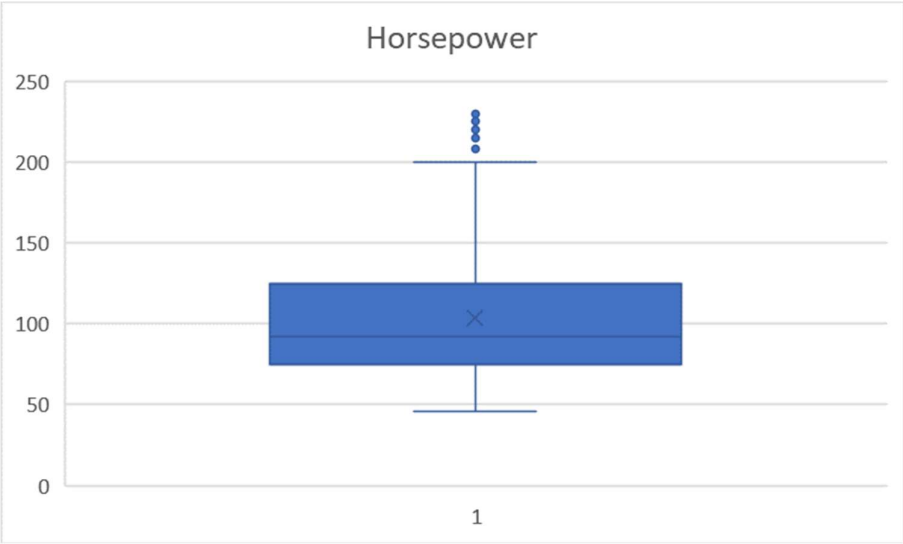

Removed the value since only one outlier found



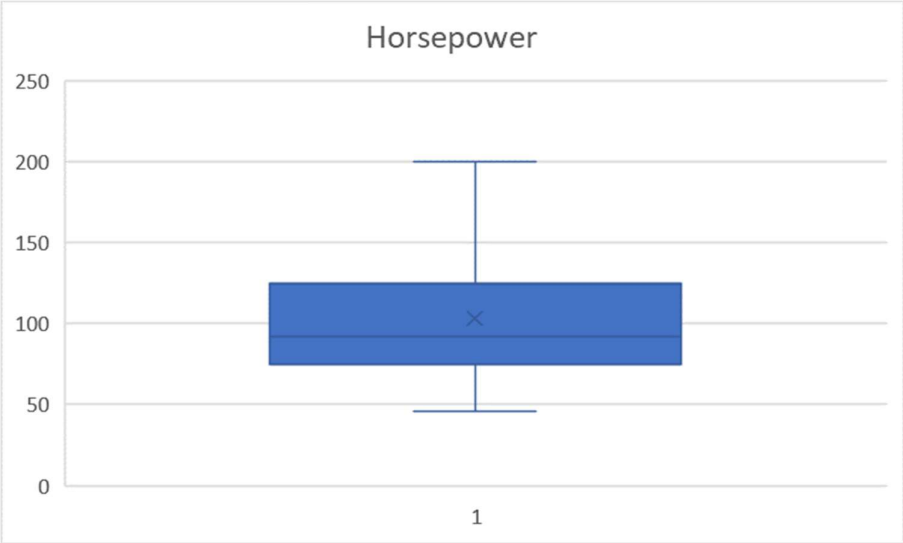

Displacement
No outlier found



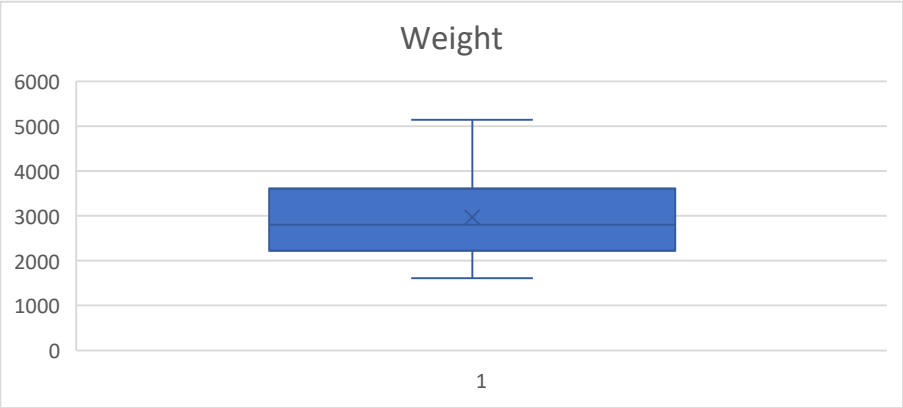

Horsepower



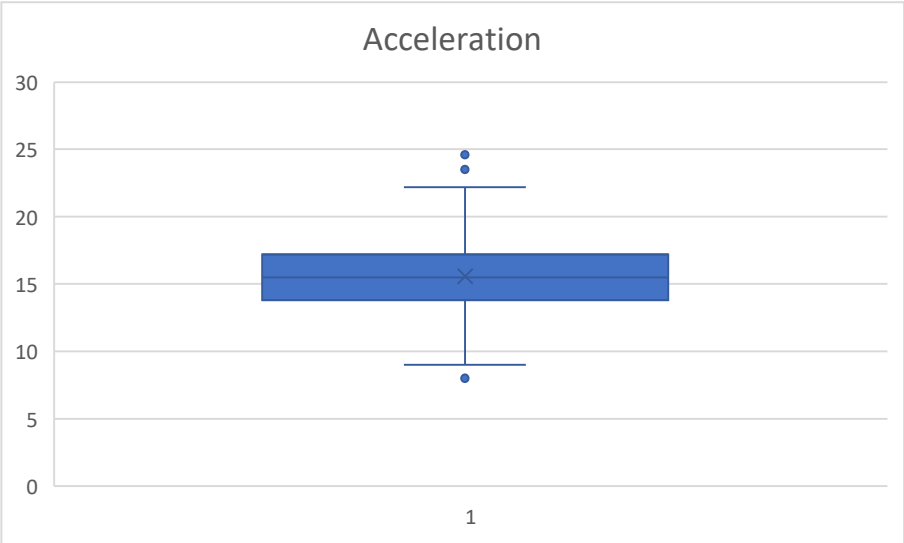

Replaced the values with high whisker value of boxplot.



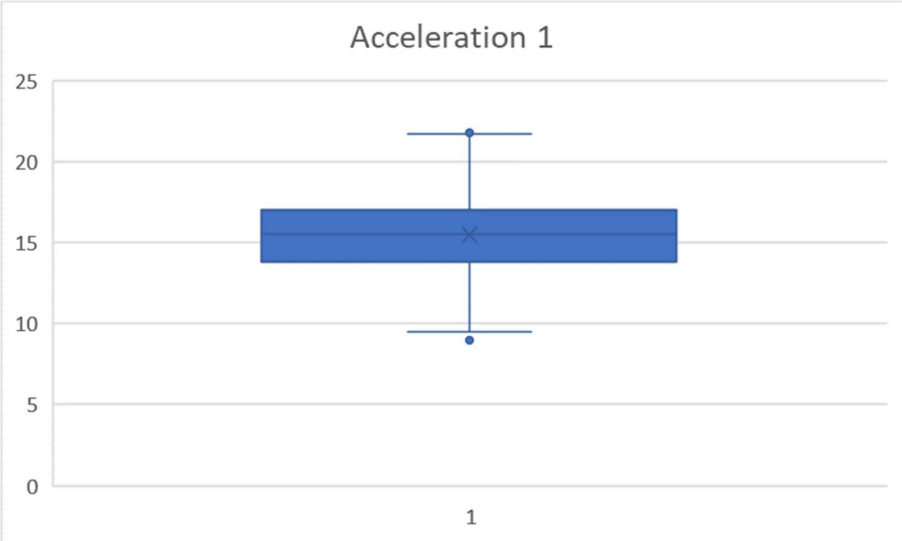

weight



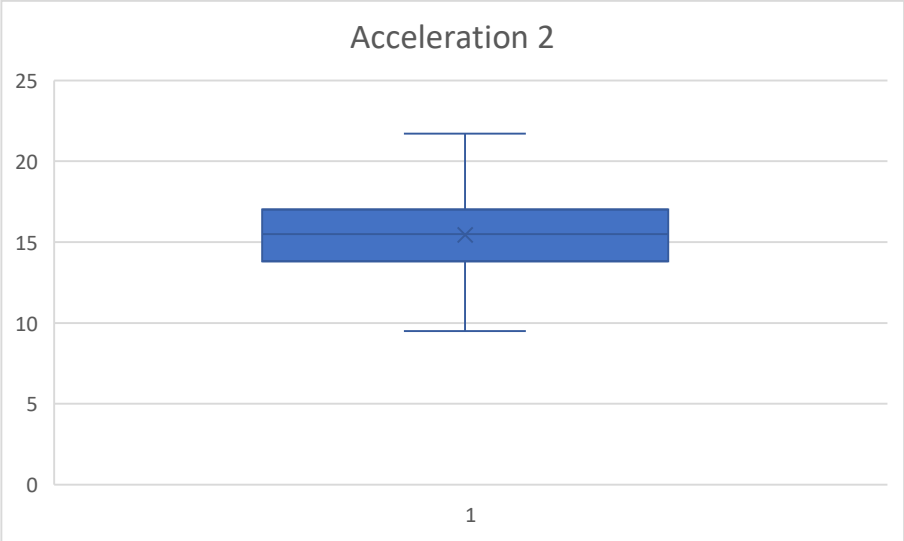

Acceleration




Removed about 10 outliers lower and above of whiskers




Again doing same process all the outliers are removed



## Correlation analysis

| | mpg | cylinders | isplacemen | orsepowe | weight | cceleratior | model year | origin | Price | ompany_nc |
|---|---|---|---|---|---|---|---|---|---|---|
| mpg | 1 | | | | | | | | | |
| cylinders | -0.78233 | 1 | | | | | | | | |
| displaceme | -0.8122 | 0.952092 | 1 | | | | | | | |
| horsepow | -0.78677 | 0.853278 | 0.89493 | 1 | | | | | | |
| weight | -0.83973 | 0.89924 | 0.936972 | 0.875253 | 1 | | | | | |
| acceleratic | 0.40409 | -0.50766 | -0.53596 | -0.69076 | -0.42957 | 1 | | | | |
| model yea | 0.561994 | -0.32861 | -0.34532 | -0.38372 | -0.2876 | 0.256266 | 1 | | | |
| origin | 0.56204 | -0.55921 | -0.61122 | -0.44775 | -0.58347 | 0.203898 | 0.164335 | 1 | | |
| Price | 0.015092 | 0.09311 | 0.076013 | 0.12243 | 0.05441 | -0.11803 | 0.029245 | -0.03805 | 1 | |
| company_ | 0.27464 | -0.28166 | -0.29831 | -0.22818 | -0.26343 | 0.095719 | 0.062688 | 0.369097 | 0.000429 | 1 |

There is high correlation seen between

1. Cylinder and displacement: therefore, removing cylinder first

| | mpg | isplacemen | orsepowe | weight | cceleratior | model year | origin | Price | ompany_nc |
|---|---|---|---|---|---|---|---|---|---|
| mpg | 1 | | | | | | | | |
| displaceme | -0.8122 | 1 | | | | | | | |
| horsepow | -0.78677 | 0.89493 | 1 | | | | | | |
| weight | -0.83973 | 0.936972 | 0.875253 | 1 | | | | | |
| acceleratic | 0.40409 | -0.53596 | -0.69076 | -0.42957 | 1 | | | | |
| model yea | 0.561994 | -0.34532 | -0.38372 | -0.2876 | 0.256266 | 1 | | | |
| origin | 0.56204 | -0.61122 | -0.44775 | -0.58347 | 0.203898 | 0.164335 | 1 | | |
| Price | 0.015092 | 0.076013 | 0.12243 | 0.05441 | -0.11803 | 0.029245 | -0.03805 | 1 | |
| company_ | 0.27464 | -0.29831 | -0.22818 | -0.26343 | 0.095719 | 0.062688 | 0.369097 | 0.000429 | 1 |

2. Again, high correlation between displacement and hp and weight therefore removing displacement

| | mpg | orsepowe | weight | cceleratior | model year | origin | Price | ompany_nc |
|---|---|---|---|---|---|---|---|---|
| mpg | 1 | | | | | | | |
| horsepow | -0.78677 | 1 | | | | | | |
| weight | -0.83973 | 0.875253 | 1 | | | | | |
| acceleratic | 0.40409 | -0.69076 | -0.42957 | 1 | | | | |
| model yea | 0.561994 | -0.38372 | -0.2876 | 0.256266 | 1 | | | |
| origin | 0.56204 | -0.44775 | -0.58347 | 0.203898 | 0.164335 | 1 | | |
| Price | 0.015092 | 0.12243 | 0.05441 | -0.11803 | 0.029245 | -0.03805 | 1 | |
| company_ | 0.27464 | -0.22818 | -0.26343 | 0.095719 | 0.062688 | 0.369097 | 0.000429 | 1 |

3. Again high correlation found between horsepower and weight removing weight

| | mpg | orsepowe | cceleratior | model year | origin | Price | ompany_nc |
|---|---|---|---|---|---|---|---|
| mpg | 1 | | | | | | |
| horsepow | -0.78677 | 1 | | | | | |
| acceleratic | 0.40409 | -0.69076 | 1 | | | | |
| model yea | 0.561994 | -0.38372 | 0.256266 | 1 | | | |
| origin | 0.56204 | -0.44775 | 0.203898 | 0.164335 | 1 | | |
| Price | 0.015092 | 0.12243 | -0.11803 | 0.029245 | -0.03805 | 1 | |
| company_ | 0.27464 | -0.22818 | 0.095719 | 0.062688 | 0.369097 | 0.000429 | 1 |

4. High negative correlation found between mpg and horsepower and mpg therefore removing mpg

| | *horsepowe* | *cceleratior* | *model year* | *origin* | *Price* | *ompany_no* |
|---|---|---|---|---|---|---|
| horsepow | 1 | | | | | |
| acceleratic | -0.69076 | 1 | | | | |
| model yea | -0.38372 | 0.256266 | 1 | | | |
| origin | -0.44775 | 0.203898 | 0.164335 | 1 | | |
| Price | 0.12243 | -0.11803 | 0.029245 | -0.03805 | 1 | |
| company_ | -0.22818 | 0.095719 | 0.062688 | 0.369097 | 0.000429 | 1 |

5.removing acceleration

| *ompany_no* | *horsepower* | *model year* | *origin* | *Price* | *company_no* |
|---|---|---|---|---|---|
| horsepow | 1 | | | | |
| model yea | -0.383724935 | 1 | | | |
| origin | -0.447747895 | 0.164335217 | 1 | | |
| Price | 0.122430384 | 0.029244601 | -0.038054871 | 1 | |
| company_ | -0.228177685 | 0.062687869 | 0.369097204 | 0.00042915 | 1 |

6.Now none of the variables has high correlation. Therefore multicollinearity is removed.

## Model Extraction:

Taking only significant variables

| | Coefficients | andard Err | t Stat | P-value | Lower 95% | Upper 95% | ower 95.0% | pper 95.0% |
|---|---|---|---|---|---|---|---|---|
| Intercept | -6.28905 | 5.652527 | -1.11261 | 0.266584 | -17.4034 | 4.825286 | -17.4034 | 4.825286 |
| company_ | 0.015885 | 0.030203 | 0.525951 | 0.59923 | -0.0435 | 0.075273 | -0.0435 | 0.075273 |
| horsepow | 0.021502 | 0.007747 | 2.77561 | 0.005784 | 0.00627 | 0.036735 | 0.00627 | 0.036735 |
| model yea | 0.113065 | 0.068937 | 1.640116 | 0.101813 | -0.02248 | 0.248614 | -0.02248 | 0.248614 |
| origin | 0.069073 | 0.338933 | 0.203795 | 0.838623 | -0.59736 | 0.735503 | -0.59736 | 0.735503 |

1.Since P value of origin is highest removing origin as attribute

| | Coefficients | andard Err | t Stat | P-value | Lower 95% | Upper 95% | ower 95.0% | pper 95.0% |
|---|---|---|---|---|---|---|---|---|
| Intercept | -6.14575 | 5.601528 | -1.09716 | 0.273269 | -17.1597 | 4.868211 | -17.1597 | 4.868211 |
| company_ | 0.017772 | 0.028713 | 0.618948 | 0.536323 | -0.03869 | 0.074229 | -0.03869 | 0.074229 |
| horsepow | 0.020908 | 0.007169 | 2.916593 | 0.003749 | 0.006813 | 0.035004 | 0.006813 | 0.035004 |
| model yea | 0.113057 | 0.06885 | 1.642071 | 0.101405 | -0.02232 | 0.248433 | -0.02232 | 0.248433 |

2. Since P value of company is highest removing company as attribute

| | Coefficients | andard Err | t Stat | P-value | Lower 95% | Upper 95% | ower 95.0% | pper 95.0% |
|---|---|---|---|---|---|---|---|---|
| Intercept | -5.69753 | 5.55001 | -1.02658 | 0.305271 | -16.6101 | 5.215043 | -16.6101 | 5.215043 |
| horsepow | 0.019926 | 0.006985 | 2.852612 | 0.004573 | 0.006191 | 0.03366 | 0.006191 | 0.03366 |
| model yea | 0.111878 | 0.068768 | 1.626894 | 0.104589 | -0.02334 | 0.247091 | -0.02334 | 0.247091 |

3. Since P value of model year is highest removing model year as attribute

| | Coefficients | andard Err | t Stat | P-value | Lower 95% | Upper 95% | ower 95.0% | pper 95.0% |
|---|---|---|---|---|---|---|---|---|
| Intercept | 3.258745247 | 0.705824 | 4.61694 | 5.33E-06 | 1.870948 | 4.646542 | 1.870948 | 4.646542 |
| horsepower | 0.015565027 | 0.006464 | 2.40786 | 0.016521 | 0.002855 | 0.028275 | 0.002855 | 0.028275 |

5. Only one significant variable can be extracted in a model. i.e. horsepower with p value 0.016521

Model equation can be given as:

Y(price) = 3.258745247 + 0.015565027*(horsepower)

## Model Evaluation:

ANOVA

| | *df* | *SS* | *MS* | *F* | *Significance F* |
|---|---|---|---|---|---|
| Regression | 1 | 120.4494 | 120.4494 | 5.797789 | 0.016520777 |
| Residual | 381 | 7915.297 | 20.77506 | | |
| Total | 382 | 8035.746 | | | |

Model is overall significant since f significance is less than 0.05

## Model diagnostic:

| | |
|---|---|
| *Regression Statistics* | |
| Multiple R | 0.122430384 |
| R Square | 0.014989199 |
| Adjusted R Square | 0.012403869 |
| Standard Error | 4.557966337 |
| Observations | 383 |

R square value of the model is 0.0149 i.e. about 1.5% of variation is target(prices) variable is explained by other significant variables.